

Tissue-resident macrophages originate from yolk-sac-derived erythro-myeloid progenitors.

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Most haematopoietic cells renew from adult haematopoietic stem cells (HSCs), however, macrophages in adult tissues can self-maintain independently of HSCs. Progenitors with macrophage potential in vitro have been described in the yolk sac before emergence of HSCs, and fetal macrophages can develop independently of Myb, a transcription factor required for HSC, and can persist in adult tissues. Nevertheless, the origin of adult macrophages and the qualitative and quantitative contributions of HSC and putative non-HSC-derived progenitors are still unclear. Here we show in mice that the vast majority of adult tissue-resident macrophages in liver (Kupffer cells), brain (microglia), epidermis (Langerhans cells) and lung (alveolar macrophages) originate from a Tie2(+) (also known as Tek) cellular pathway generating Csf1r(+) erythro-myeloid progenitors (EMPs) distinct from HSCs. EMPs develop in the yolk sac at embryonic day (E) 8.5, migrate and colonize the nascent fetal liver before E10.5, and give rise to fetal erythrocytes, macrophages, granulocytes and monocytes until at least E16.5. Subsequently, HSC-derived cells replace erythrocytes, granulocytes and monocytes. Kupffer cells, microglia and Langerhans cells are only marginally replaced in one-year-old mice, whereas alveolar macrophages may be progressively replaced in ageing mice. Our fate-mapping experiments identify, in the fetal liver, a sequence of yolk sac EMP-derived and HSC-derived haematopoiesis, and identify yolk sac EMPs as a common origin for tissue macrophages.

Interaction of extraembryonic microglia and neonatal brain development.

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Microglia are resident immune cells in the central nervous system (CNS), which, in a healthy state, promote CNS homeostasis and respond to CNS injury. In contrast, microglia are also implicated in pathological conditions where they may contribute to neural damage. Primitive microglia arise from extraembryonic progenitors in the yolk sac (YS). The extraembryonic origins of primitive microglia are distinct from other tissue macrophages. The YS is the first site of hematopoiesis in development. Uniquely, microglial pregenital cells in the mouse derive from an early myeloid branch of the hematopoietic lineage in the YS. Microglia are critical in several key stages of physiological brain development, including embryonic vasculogenesis, immunosurveillance, and neurogenesis. Abnormal microglial function has been linked to neurodevelopmental and neurodegenerative diseases, although mechanistic roles in disease etiology remain incompletely understood. Knowledge of species-specific differences between human, murine and other animal models is also critical to understanding translational relevance to human health and disease as biomedical understanding of the importance of primitive microglia advances. This significance drives the importance of understanding, comparatively, the extraembryonic origins and developmental mechanisms whereby human primitive microglia differentiate and migrate to inform translational research. A better understanding of the molecular drivers may lead to biomarkers and/or preventative or therapeutic measures for neonatal brain development and neurodegenerative diseases. Herein, the role of microglia in neonatal brain development is discussed, current understandings of the developmental origins of microglia are described, the ontogeny and phylogeny of microglia, and implications of in vitro microglia-like cell differentiation, with a specific interest on neurodegenerative diseases, are reviewed. Together, these emphasize the importance of leveraging the extraembryonic origins of microglia to not only better understand neurodevelopment and neurodegenerative diseases, but also to develop protective measures

that are specific to human microglia.

Microglia are effector cells of CD47-SIRP α antiphagocytic axis disruption against glioblastoma.

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Glioblastoma multiforme (GBM) is a highly aggressive malignant brain tumor with fatal outcome. Tumor-associated macrophages and microglia (TAMs) have been found to be major tumor-promoting immune cells in the tumor microenvironment. Hence, modulation and reeducation of tumor-associated macrophages and microglia in GBM is considered a promising antitumor strategy. Resident microglia and invading macrophages have been shown to have distinct origin and function. Whereas yolk sac-derived microglia reside in the brain, blood-derived monocytes invade the central nervous system only under pathological conditions like tumor formation. We recently showed that disruption of the SIRP α -CD47 signaling axis is efficacious against various brain tumors including GBM primarily by inducing tumor phagocytosis. However, most effects are attributed to macrophages recruited from the periphery but the role of the brain resident microglia is unknown. Here, we sought to utilize a model to distinguish resident microglia and peripheral macrophages within the GBM-TAM pool, using orthotopically xenografted, immunodeficient, and syngeneic mouse models with genetically color-coded macrophages (Ccr2RFP) and microglia (Cx3cr1GFP). We show that even in the absence of phagocytizing macrophages (Ccr2RFP/RFP), microglia are effector cells of tumor cell phagocytosis in response to anti-CD47 blockade. Additionally, macrophages and microglia show distinct morphological and transcriptional changes. Importantly, the transcriptional profile of microglia shows less of an inflammatory response which makes them a promising target for clinical applications.

Genetic targeting of microglia.

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Genetic targeting of microglia and other myeloid cells in the central nervous system (CNS) is highly desirable as they are critical effectors and regulators of changes in CNS homeostasis during development as well as in health and disease. Therefore, genetic reprogramming of microglia could constitute a central approach for potentially reducing disease burden. Previous attempts to target only microglia in vivo failed because of the similarities to their hematopoietic relatives in the circulation. However, this concept has been challenged by recent results of developmental and gene expression profiling studies which used novel molecular biological tools to unravel the origin of microglia and to define their role as specialized tissue macrophages clearly distinct from monocytes or monocyte-derived macrophages. The aim of this review is to recapitulate the history of microglia targeting approaches and finally highlight recent achievements in the field. We will discuss the pros and cons of the newly available genetic tools, their potential for future microglia research and genetic strategies.

The niche matters: origin, function and fate of CNS-associated macrophages during health and disease.

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There are several cellular and acellular structural barriers associated with the brain interfaces, which include the dura, the leptomeninges, the perivascular space and the choroid plexus epithelium. Each structure is enriched by distinct myeloid populations, which mainly originate from erythromyeloid precursors (EMP) in the embryonic yolk sac and seed the CNS during embryogenesis. However, depending on the precise microanatomical environment, resident myeloid cells differ in their marker profile, turnover and the extent to which they can be replenished by blood-derived cells. While some EMP-derived cells seed the parenchyma to become microglia, others engraft the meninges and become CNS-associated macrophages (CAMs), also referred to as border-associated macrophages (BAMs), e.g., leptomeningeal macrophages (MnM Φ). Recent data revealed that MnM Φ migrate into perivascular spaces postnatally where they differentiate into perivascular macrophages (PvM Φ). Under homeostatic conditions in pathogen-free mice, there is virtually no contribution of bone marrow-derived cells to MnM Φ and PvM Φ , but rather to macrophages of the choroid plexus and dura. In neuropathological conditions in which the blood-brain barrier is compromised, however, an influx of bone marrow-derived cells into the CNS can occur, potentially contributing to the pool of CNS myeloid cells. Simultaneously, resident CAMs may also proliferate and undergo transcriptional and proteomic changes, thereby, contributing to the disease outcome. Thus, both resident and infiltrating myeloid cells together act within their microenvironmental niche, but both populations play crucial roles in the overall disease course. Here, we summarize the current understanding of the sources and fates of resident CAMs in health and disease, and the role of the microenvironment in influencing their maintenance and function.

Single-cell RNA sequencing-guided fate-mapping toolkit delineates the contribution of yolk sac erythro-myeloid progenitors.

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Erythro-myeloid progenitors of the yolk sac that originates during early embryo development has been suggested to generate tissue-resident macrophage, mast cell, and even endothelial cell populations from fetal to adult stages. However, the heterogeneity of erythro-myeloid progenitors (EMPs) is not well characterized. Here, we adapt single-cell RNA sequencing to dissect the heterogeneity of EMPs and establish several fate-mapping tools for each EMP subset to trace the contributions of different EMP subsets. We identify two primitive and one definitive EMP subsets from the yolk sac. In addition, we find that primitive EMPs are decoupled from definitive EMPs. Furthermore, we confirm that primitive and definitive EMPs give rise to microglia and other tissue-resident macrophages, respectively. In contrast, only Kit⁺ Csf1r⁻ primitive EMPs generate endothelial cells transiently during early embryo development. Overall, our results delineate the contribution of yolk sac EMPs more clearly based on the single-cell RNA sequencing (scRNA-seq)-guided fate-mapping toolkit.

Distinguishing the effects of systemic CSF1R inhibition by PLX3397 on microglia and peripheral immune cells.

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Microglia, the primary immune cells of the central nervous system (CNS), are derived from the yolk sac and populate the brain during development. Once microglia migrate to the CNS, they are self-renewing

and require CSF1R signaling for their maintenance. Pexidartinib (PLX3397, PLX), a small molecule inhibitor of the CSF1R, has been shown to effectively deplete microglia since microglial maintenance is CSF1R-dependent. There have, however, been several conflicting reports that have shown the potential off-target effects of PLX on peripheral immune cells particularly those of lymphoid origin. Given this controversy in the use of the PLX family of drugs, it has become important to ascertain to what extent PLX affects the peripheral immune profile in lymphoid (spleen, and bone marrow) and non-lymphoid (kidney, lungs, and heart) organs. PLX3397 chow treatment at 660 mg/kg for 7 days significantly reduced CD45+ macrophages, CX3CR1-GFP cells, CD11b+CD45intermediate cells, and P2RY12 expression in the brain. However, there were minimal effects on peripheral immune cells from both lymphoid and non-lymphoid organs except in the heart where there was a significant decrease in CD3+ cells, inflammatory and patrolling monocytes, and CD11b+Ly6G+ neutrophils. We then stimulated the immune system with 1 mg/kg of LPS which resulted in a significant reduction in the number of innate immune cells. In this context, PLX did not alter the cytokine profile in the serum and the brain of naïve mice but did so in the LPS-stimulated group resulting in a significant reduction in TNF α , IL-1 α , IFN- γ and IL-1 β . Furthermore, PLX did not alter locomotor activity in the open field test suggesting that microglia do not contribute to LPS-induced sickness behavior. Our results provide an assessment of immune cell populations with PLX3397 treatment on brain, lymphoid and non-lymphoid organs without and during LPS treatment that can serve as a resource for understanding consequences of such approaches.

Multiple mechanisms of microglia: a gatekeeper's contribution to pain states.

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Microglia are gatekeepers in the CNS for a wide range of pathological stimuli and they blow the whistle when things go wrong. Collectively, microglia form a CNS tissue alarm system (Kreutzberg's "sensor of pathology"), and their involvement in physiological pain is in line with this function. However, pathological neuropathic pain is characterized by microglial activation that is unwanted and considered to contribute to or even cause tactile allodynia, hyperalgesia and spontaneous pain. Such abnormal microglial behavior seems likely due to an as yet ill-understood disturbance of microglial functions unrelated to inflammation. The idea that microglia have roles in the CNS that differ from those of peripheral macrophages has gained momentum with the discovery of their separate, pre-hematopoietic lineage during embryonic development and their direct interactions with synapses.

Analysis of Microglia and Monocyte-derived Macrophages from the Central Nervous System by Flow Cytometry.

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Numerous studies have demonstrated the role of immune cells, in particular macrophages, in central nervous system (CNS) pathologies. There are two main macrophage populations in the CNS: (i) the microglia, which are the resident macrophages of the CNS and are derived from yolk sac progenitors during embryogenesis, and (ii) the monocyte-derived macrophages (MDM), which can infiltrate the CNS during disease and are derived from bone marrow progenitors. The roles of each macrophage subpopulation differ depending on the pathology being studied. Furthermore, there is no consensus on the histological markers or the distinguishing criteria used for these macrophage subpopulations.

However, the analysis of the expression profiles of the CD11b and CD45 markers by flow cytometry allows us to distinguish the microglia (CD11b+CD45med) from the MDM (CD11b+CD45high). In this protocol, we show that the density gradient centrifugation and the flow cytometry analysis can be used to characterize these CNS macrophage subpopulations, and to study several markers of interest expressed by these cells as we recently published. Thus, this technique can further our understanding of the role of macrophages in mouse models of neurological diseases and can also be used to evaluate drug effects on these cells.

Fate mapping reveals that microglia and recruited monocyte-derived macrophages are definitively distinguishable by phenotype in the retina.

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The recent paradigm shift that microglia are yolk sac-derived, not hematopoietic-derived, is reshaping our knowledge about the isolated role of microglia in CNS diseases, including degenerative conditions of the retina. However, unraveling microglial-specific functions has been hindered by phenotypic overlap of microglia with monocyte-derived macrophages. The latter are differentiated from recruited monocytes in neuroinflammation, including retina. Here we demonstrate the use of fate mapping wherein microglia and monocyte-derived cells are endogenously labeled with different fluorescent reporters. Combining this method with 12-color flow cytometry, we show that these two populations are definitively distinguishable by phenotype in retina. We prove that retinal microglia have a unique CD45(lo) CD11c(lo) F4/80(lo) I-A/I-E(-) signature, conserved in the steady state and during retinal injury. The latter was observed in the widely used light-induced retinal degeneration model and corroborated in other models, including whole-body irradiation/bone-marrow transplantation. The literature contains conflicting observations about whether microglia, including in the retina, increase expression of these markers in neuroinflammation. We show that monocyte-derived macrophages have elevated expression of these surface markers, not microglia. Our resolution of such phenotypic differences may serve as a robust way to help characterize isolated roles of these cells in retinal neuroinflammation and possibly elsewhere in CNS.
